

Software Requirements Specification
Restaurant App MVP (Frontend Only, React Native)
— Fall 2026

Assignment No. 1

1. Introduction

Purpose: define a frontend-only React Native MVP using mock/local data, React hooks, Context, useReducer and local persistence concepts. Scope includes customer ordering, reservations, tracking, manager functions and theme. Out of scope: real payments, backend/database services, push notifications, external APIs and production authentication.

Requirement note: every FR is intended to be implemented as a client-side behavior in the MVP.

2. Overall Description

Roles: Customer and Restaurant Manager. Customer stories include sign in, browse categories, search, favourite items, add to cart, apply promo, reserve a table, place an order, track status and switch theme. Manager stories include view incoming orders, update status, manage reservations and manage menu availability/prices.

Requirement note: every FR is intended to be implemented as a client-side behavior in the MVP.

3. Functional Requirements

Authentication: FR-01 login; FR-02 signup; FR-03 confirm password; FR-04 password validation; FR-05 loading state; FR-06 role routing. Menu: FR-07 categories; FR-08 menu items; FR-09 availability; FR-10 specials; FR-11 count; FR-12 refresh; FR-13 favourite; FR-14 sorting. Search: FR-15 search; FR-16 400ms debounce; FR-17 clear/focus; FR-18 recent searches; FR-19 empty state; FR-20 back-to-top; FR-21 previous-query prevention. Cart: FR-22 add; FR-23 quantity; FR-24 notes/promo; FR-25 live total. Reservation: FR-26 date; FR-27 time; FR-28 party 1–12; FR-29 Pakistani mobile format; FR-30 one-hour-ahead rule; FR-31 available tables; FR-32 confirmation; FR-33 cancellation. Orders: FR-34 order type; FR-35 status progression; FR-36 elapsed/progress; FR-37 manager override. Manager: FR-38 incoming orders; FR-39 reservations; FR-40 menu management.

Requirement note: every FR is intended to be implemented as a client-side behavior in the MVP.

4. Non-Functional Requirements

Usability: clear controls and validation. Performance: responsive local lists and simulated loading. Responsive design: common phone screen sizes. Maintainability: reusable data, reducers and hooks. Data handling: local mock data with no backend dependency.

Requirement note: every FR is intended to be implemented as a client-side behavior in the MVP.

5. Client-Side Data Model

Users, Categories, Menuitems, Tables, Reservations, Orders and Promo Codes are represented as local JavaScript data. React state/context/reducers provide runtime state.

Requirement note: every FR is intended to be implemented as a client-side behavior in the MVP.

6. UML Diagrams

The five diagrams are included in A1/UML: Use Case, Class, Sequence, State Machine and Component diagrams.

Requirement note: every FR is intended to be implemented as a client-side behavior in the MVP.

7. MVP Frontend

Login/Signup; Menu Browsing; Search and Scroll Controls; Profile and Theme; Cart; Order Summary; Table Reservation; Order Tracking and Manager Dashboard. The implementation demonstrates useState, useEffect, useRef, useContext, useReducer, useMemo, useCallback and custom hooks.

Requirement note: every FR is intended to be implemented as a client-side behavior in the MVP.